

- Suppose the Godel number is 300. Divide by 2 as many times as possible. It is possible to do it twice. Hence first number in this sequence is 2. Then we get 75. 3 divides 75 once. The second number is 3 and we get 25. 5 divides it twice. So the third number in the sequence is 2 and we get 2, 1, 2.
- Giving numbers for alphabets like 1 for 'a', 2 for 'b', etc. a word and hence a sentence can get a Godel numbering. It should be noted that for sentences the Godel number will be very large.
- Godel constructed a statement somewhat similar to the one given below.
- Statement 1:

 The statement whose Godel numbering is x , is not a theorem.
(x was specified in a particular manner, using a concept called arithmoquintification).
- When the Godel number of this statement 1 was calculated, it was found to be x .
- If there exists an axiomatic complete and consistent system, every true statement must be a theorem there.
- Suppose there exists such a system. Then if statement 1 is derived from the axioms using the rules of inference, it is a theorem. But statement 1 says it is not a theorem. Hence we arrive at a contradiction. If statement 1 cannot be derived from the axioms using rules of inference, it is not a theorem and statement 1 is a true statement which cannot be proved in the axiomatic system, i.e. the system is not complete.
- Thus Godel stated that for any sound logical axiomatic system that is sufficiently rich to contain the theory of numbers, there must be statements that can neither be proved nor disproved.

Some further reading

1. The nuts and bolts of proofs: Antonella Cupillari
2. Bridge to Abstract Mathematics: R.P. Morash
3. How to Read and Do Proofs: An Introduction to Mathematical Thought Processes: Daniel Solow